

Orbital Collapse Architecture: A Power-Law Model of Institutional Absorption Following Systemic Financial Failure

Elán Zainos Corona · Fractal Core Research, Tlaxcala, Mexico · ORCID 0009-0009-9125-253X

Prepared for the MIT Golub Center for Finance and Policy 13th Annual Conference — "Financial Regulation in an Era of Innovation and Disruption," October 29–30, 2026.

Working paper — preliminary draft. Data & code: github.com/Inzainos/The-shadow-Node-Theory

Abstract

When a dominant financial institution fails, its assets, counterparties, and market position are not destroyed — they are *absorbed* by an identifiable successor, and the speed of that absorption is empirically regular. We model the post-failure dominance ratio $R(t) = \text{mass}_{\text{absorber}}(t) / \text{mass}_{\text{collapsed, peak}}$ as a power law, $R(t) = a \cdot t^b$, and interpret the exponent **b** as the velocity of institutional absorption following functional extinction of a failed hub.

Applying this Orbital Collapse Architecture (ACO) framework to the six canonical absorptions of the 2008–2009 crisis — Lehman Brothers, Bear Stearns, Washington Mutual, Wachovia, Merrill Lynch, and Chrysler — we recover statistically significant power-law absorption curves in every case ($R^2 = 0.85\text{--}0.99$, all $p < 0.01$), reconstructed entirely from primary sources. The central regulatory observation is that **the resolution regime is legible in the absorption exponent**: regulator-brokered resolutions (Washington Mutual via FDIC, Bear Stearns via the Federal Reserve) exhibit near-instant absorption ($b \rightarrow 0$), whereas the single disorderly bankruptcy in the sample (Lehman Brothers) shows the slowest absorption (highest b). We position these signatures as a quantitative, falsifiable lens on too-big-to-fail concentration and resolution policy.

1. Introduction

The 2008–2009 financial crisis produced a wave of institutional failures whose common structural feature has been under-theorized: in nearly every case, a failing institution's balance sheet, client relationships, and market position did not evaporate but were *captured* by an identifiable successor. Bear Stearns became part of JPMorgan; Merrill Lynch became part of Bank of America; Washington Mutual's banking operations passed, within 48 hours, to JPMorgan under FDIC receivership. The crisis was, in large part, an episode of accelerated **concentration** — the very dynamic that too-big-to-fail regulation is meant to constrain.

This paper asks a simple quantitative question: *how fast* does absorption happen, and *what governs that speed*? We argue that the post-failure dominance ratio between absorber and absorbed follows a power law, and that the fitted exponent carries policy-relevant information about the resolution regime under which the failure was processed.

The framework is a financial-sector application of a more general model — Shadow Node Theory (SNT) — which characterizes how dominance evolves between a coupled hub and node across domains as diverse as demography, epidemiology, and astronomy. The financial cases analyzed here are one domain-specific slice of a larger 721-case cross-domain validation, which we draw on only to establish that the mechanism is not financial-sector-specific.

2. Model

Let the absorber's captured mass at time t after the trigger be $M_A(t)$, and let the failed institution's peak pre-failure mass be M_C . Define the absorption ratio $R(t) = M_A(t) / M_C$. We posit that $R(t)$ follows a power law in time since the trigger,

$$R(t) = a \cdot t^b \quad (\text{Eq. 1})$$

estimated by ordinary least squares on log-transformed axes, $\log R = \log a + b \cdot \log t$. The exponent b is the object of interest: $b \rightarrow 0$ denotes near-instantaneous absorption — the successor captures the bulk of the mass almost immediately, then plateaus (characteristic of brokered or pre-arranged resolutions); a **moderate** $b > 0$ denotes gradual absorption over an extended window, consistent with contested, market-driven, or litigation-encumbered transfers.

The ACO criterion (definitional). A case qualifies as Orbital Collapse Architecture only if both (1) the hub undergoes *functional extinction* and (2) its resources are absorbed by a *specific, identifiable* node. Failures that dissolve without capture (e.g., outright liquidation) are excluded; this distinguishes ACO from generic decline.

3. Data

All six cases are reconstructed from primary regulatory and corporate records: SEC EDGAR filings (8-K, S-4, 10-K), the Federal Reserve Flow of Funds, FDIC failure and acquisition records, SIGTARP reports, the U.S. Treasury TARP tracker, and the Valukas Lehman Examiner Report. No values are synthetic. Time series are reconstructed at monthly resolution from the trigger event (the failure or forced-sale date) forward.

Hub → Absorber	Resolution channel	Trigger
Lehman Brothers → Barclays + JPMorgan	Disorderly bankruptcy (Ch. 11)	Sep 2008
Bear Stearns → JPMorgan	Fed-brokered (Maiden Lane)	Mar 2008
Washington Mutual → JPMorgan	FDIC receivership (P&A)	Sep 2008
Wachovia → Wells Fargo	Open-bank, FDIC-adjacent	Oct 2008
Merrill Lynch → Bank of America	Negotiated merger	Sep 2008
Chrysler → Fiat + US Treasury	Govt-managed (TARP/\$363)	2009

4. Results

Every case yields a statistically significant power-law absorption curve, ranked here by absorption velocity (lowest b = fastest absorption):

Hub → Absorber	b	R^2	p
Washington Mutual → JPMorgan (FDIC)	+0.009	0.946	0.001
Bear Stearns → JPMorgan (Fed)	+0.043	0.926	0.002
Chrysler → Fiat + Treasury	+0.138	0.990	<0.001
Wachovia → Wells Fargo	+0.153	0.892	0.004
Merrill Lynch → Bank of America	+0.217	0.846	0.005
Lehman Brothers → Barclays + JPMorgan	+0.246	0.892	<0.001

The ordering is the finding. Ranked by absorption velocity, the sequence tracks the degree of regulatory pre-arrangement of each resolution. The two fastest absorptions are the two most actively brokered failures: Washington Mutual ($b = 0.009$), resolved by the FDIC in a purchase-and-assumption completed within 48 hours, and Bear Stearns ($b = 0.043$), resolved through a Federal Reserve-backed sale to JPMorgan. The slowest absorption is Lehman Brothers ($b = 0.246$), the one failure allowed to proceed as a disorderly bankruptcy. Government-managed and negotiated-merger cases fall in between. In short, the absorption exponent is a readout of resolution policy: the more orderly and pre-arranged the resolution, the closer b is to zero.

5. Discussion — regulatory implications

5.1 A quantitative signature of too-big-to-fail consolidation. Each curve documents mass flowing toward an already-large successor (JPMorgan absorbed both Bear Stearns and Washington Mutual; Bank of America absorbed Merrill). ACO gives a single number — b — summarizing how quickly that concentration completed, adding a temporal dimension to static concentration metrics.

5.2 Resolution design is legible after the fact — and potentially ex ante. If absorption velocity reflects the resolution channel, then b is a candidate ex post diagnostic for whether orderly-resolution mechanisms (Title II of Dodd-Frank, FDIC single-point-of-entry) achieve their stated aim of rapid, predictable transfer. A well-functioning regime should produce low- b absorptions; a regime that lets failures become disorderly should produce high- b ones.

5.3 Genuine absorption vs. obsolescence-without-capture. The ACO criterion excludes failures where no successor captures the mass. An absorbed firm transmits its risk and relationships into the successor (raising concentration and interconnectedness); a merely obsolete firm does not. The framework provides an empirical test for which is occurring.

6. Robustness and falsifiability

The model is stated to be refutable. We pre-commit to three refutation conditions (RC-ACO): **RC-ACO-1** — refuted for a case if the absorber's mass does not increase post-absorption (R does not grow); **RC-ACO-2** — refuted if b shows no systematic relationship to resolution channel across an expanded sample; **RC-ACO-3** — refuted if a vestigial-successor recovery ($R > 1$ over the long run) fails to appear where predicted.

Generality. The financial cases are one domain of a 721-case corpus in which institutional friction is the dominant predictor of the satellization exponent (Spearman $\rho = -0.68$, $p = 2.5 \times 10^{-97}$, $n = 714$). That friction-velocity relationship is the cross-domain analogue of the resolution-channel ordering documented here: more institutional friction (more regulatory intervention) slows the dominance dynamic. The financial results are therefore a sectoral instance of a general pattern, not an isolated curve-fit.

7. Limitations

This is a preliminary draft. (i) The sample is six cases; the resolution-channel ordering is descriptive, not yet a powered statistical test. (ii) Each $R(t)$ series is reconstructed at coarse (monthly) resolution and the mass proxy varies by case; per-case sources are reported. (iii) Absorption curves are sensitive to the choice of trigger date. A full paper will expand the sample (S&L era, European 2011–2012,

regional-bank failures of 2023), formalize the resolution-channel variable and test b against it, and add explicit comparison to the too-big-to-fail and orderly-resolution literatures.

8. Conclusion

Institutional failure in finance is, structurally, a transfer of dominance to an identifiable successor at an empirically regular rate. That rate — the absorption exponent b — is legible, reproducible from primary sources, and appears to encode the resolution regime under which a failure was processed. For a conference on financial regulation in an era of disruption, ACO offers a compact, falsifiable instrument for asking whether our resolution machinery actually delivers the orderly, rapid transfers it promises.

Appendix — data sources by case

- **Lehman Brothers:** Valukas A.R. (2010), Lehman Brothers Holdings Inc. Chapter 11 Examiner Report, U.S. Bankruptcy Court SDNY; Federal Reserve Flow of Funds 2008–2013; SEC EDGAR 10-K, Barclays & JPMorgan 2008–2009.
- **Bear Stearns:** Federal Reserve Bank of New York (2008), Maiden Lane transaction; SEC Form 8-K, JPMorgan Chase, Mar 2008; Sorkin (2009), Too Big To Fail.
- **Washington Mutual:** FDIC (2008), WaMu failure & acquisition press release, Sep 25 2008; JPMorgan Chase Form 8-K, Sep 2008; FDIC Failed Bank List.
- **Wachovia:** SEC Form 8-K, Wells Fargo, Oct 2008; FDIC statement, Oct 3 2008; Wessel (2009), In Fed We Trust.
- **Merrill Lynch:** SEC Form S-4, Bank of America, Dec 2008 (merger proxy); BofA 10-K 2009; Lewis testimony to the FCIC (2010).
- **Chrysler:** SIGTARP (2012), Chrysler Bailout Report; Chrysler Group LLC 10-K 2010; U.S. Treasury TARP tracker; Rattner (2010), Overhaul.

Full corpus, code, and reproduction instructions: github.com/lnzainos/The-shadow-Node-Theory